Toll-Like Receptor Signaling in *Pocillopora damicornis*

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https://corals.cs.tufts.edu

ABSTRACT

Toll-like receptors (TLRs) play a crucial role in innate immunity. These membrane glycoprotein receptors recognize and respond to a variety of microbial components such as lipopeptides, peptidoglycans, lipopolysaccharide, flagellin, DNA, and RNA from viruses or bacteria. Viruses and bacteria are the most abundant biological entries in seawater and live in close association with corals. The chemical crosstalk between corals and microbes plays an important role in coral growth and development. In the present study, we used our remote homology detection pipeline to find human TLR protein homologs in *Pocillopora damicornis*. Our results showed the presence of 13 unique TLR homologs in the coral *P. damicornis*. Further, using leucine rich sequence analysis and ligand active site structural analysis, we have provided the significant preliminary evidence for the presence of TLR homologs in *P. damicornis*. Our results provide valuable insights for the presence of a comprehensive **innate immunity** in corals. This research may provide valuable information to understand the coral immune system and the evolution of innate immunity in humans over the course of evolution.

RESEARCH PIPELINE

1. HHBlits protein homology system web portal built & predictions made for the 3 key coral capabilities.
2. HHblits protein homology grouping with humans and non human groups.
3. PROSITE for describing protein domains, families and functional sites as well as associated patterns and profiles to identify them

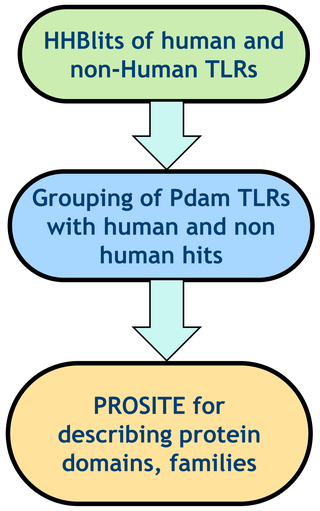

Fig.1. Pipeline for the discovery of TLRs in P. damicornis.

Corals may have 13 unique TLRs

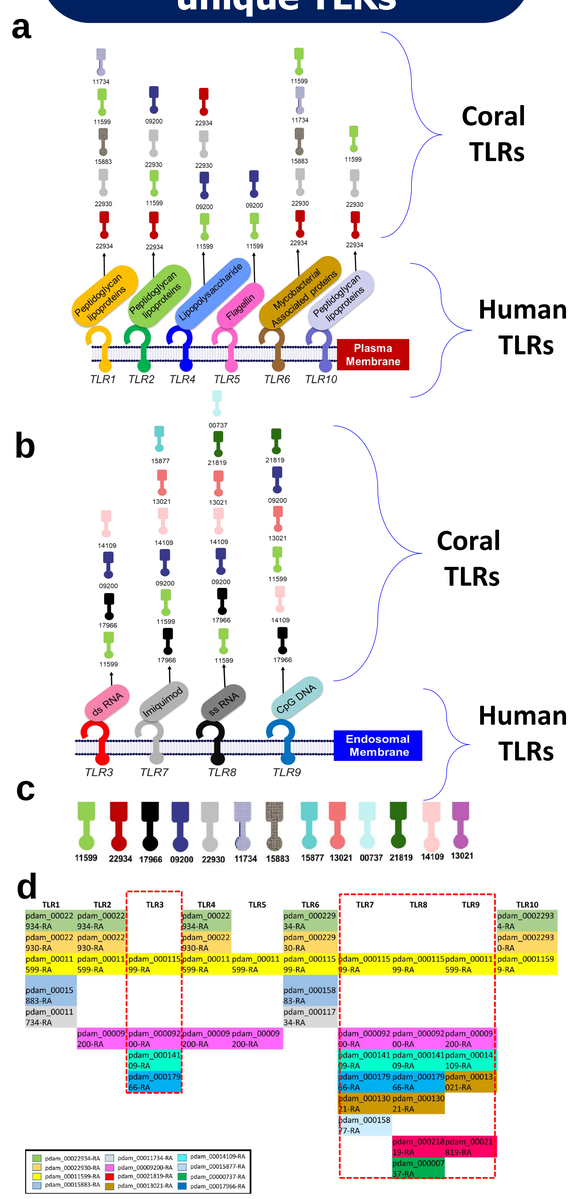

Fig.2. Schematic representation of HHBlits results. (a) grouping of p.dam hits with human TLRs present on plasma membrane (b) grouping of p.dam hits with human TLRs present on endosomal membrane (c) complete list of P.dam proteins matched with human TLRs (13) (d) Table showing the color coding match for p.dam vs human TLR proteins.

Coral showed high similarity with non-human TLRs

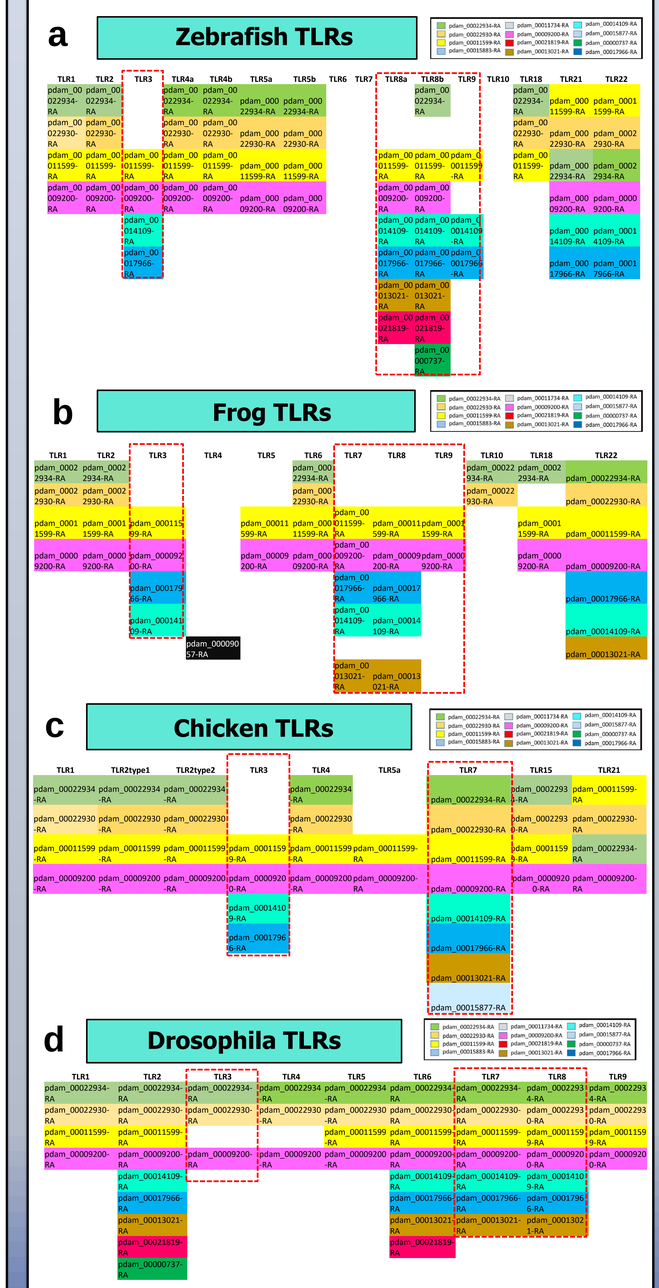

Fig.3. Color coded representation of p.dam TLR hits with non human TLR proteins (a) Zebrafish TLRs (b) frog TLRs (c) Chicken TLRs (d) Drosophila TLRs

Coral TLRs have leucine rich repeats and TIR domains

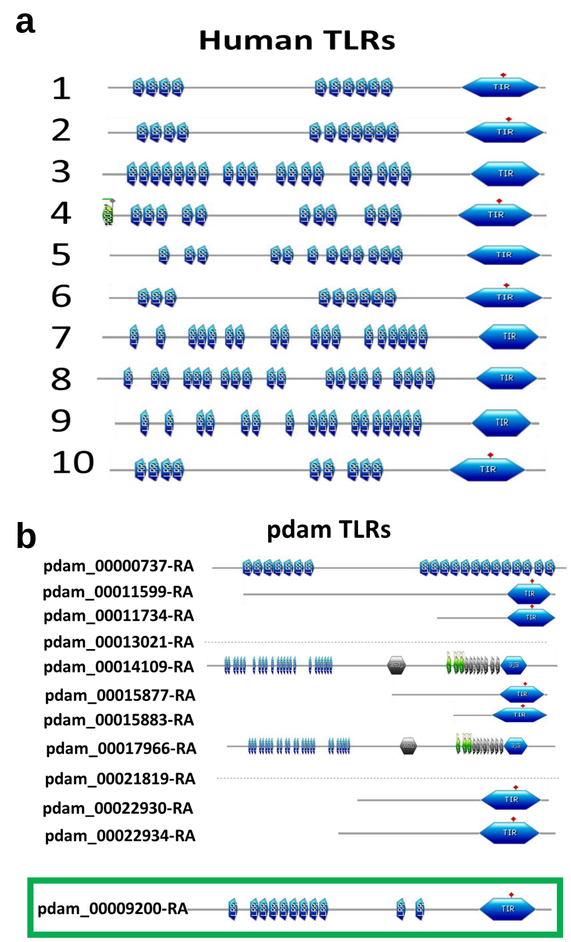

Fig.4. PROSITE results of human vs p.dam TLRs. (a) description of leucine rich repeats and TIR domain of human TLRs (b) description protein domains of p.dam TLRs. Green Rectangle: Single P. dam hit that showed leucine rich domain and TIR domain with active site (pdam_00009200)

ACKNOWLEDGEMENTS

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